

SKOVORODINO, Russian Federation

WMO: 306920

Lat: 54.000N

Lon: 123.967E

Elev: 397

StdP: 96.64

Time Zone: 9.00 (E09)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.1	(c) -38.2	(d) -42.1	(e) 0.1	(f) -39.8	(g) -40.2	(h) 0.1	(i) -38.0	(j) 7.7	(k) -17.3	(l) 6.4	(m) -19.9	(n) 1.3	(o) 50	(p) 0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	29.0	19.7	27.2	18.6	25.5	18.0	21.3	26.1	20.3	24.8	19.4	23.5	2.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	15.3	23.5	18.9	14.4	22.4	17.8	13.4	21.4	64.3	25.9	60.7	24.9	57.2	23.5	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.9	(b) 6.4	(c) 6.0	(e) -43.1	(f) 32.7	(g) 1.7	(h) 1.7	(i) -44.3	(j) 33.9	(k) -45.3	(l) 34.9	(m) -46.3	(n) 35.8	(o) -47.5	(p) 37.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.4	-26.5	-22.5	-13.4	-0.5	8.6	16.0	18.5	15.3	8.0	-2.9	-17.0	-25.8
	DBStd	16.96	5.02	4.46	5.77	5.01	3.83	3.44	2.80	3.43	4.25	5.65	6.87	5.59
	HDD10.0	5567	1131	910	724	315	71	2	0	3	88	401	811	1111
	HDD18.3	8002	1389	1143	983	564	301	86	33	104	311	659	1061	1369
	CDD10.0	668	0	0	0	1	29	181	262	168	27	1	0	0
	CDD18.3	63	0	0	0	0	1	15	37	11	0	0	0	0
	CDH23.3	900	0	0	0	0	26	298	391	177	9	0	0	0
	CDH26.7	203	0	0	0	0	3	69	94	37	0	0	0	0
Wind	WSAvg	2.3	1.8	2.1	2.5	2.9	3.0	2.5	2.1	2.1	2.3	2.2	1.9	1.8
Precipitation	PrecAvg	434	4	4	7	21	37	76	111	85	58	16	10	6
	PrecMax	545	13	9	20	54	81	197	174	146	159	47	28	14
	PrecMin	325	0	0	2	2	2	20	48	18	13	1	2	0
	PrecStd	67	3	3	5	13	24	44	40	34	38	11	6	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.8	-5.2	6.2	17.4	26.3	31.0	31.6	30.0	24.7	14.4	-0.5	-7.6
		MCWB	-12.2	-7.7	1.0	8.1	14.0	20.7	20.7	20.7	16.4	7.8	-3.1	-9.4
	2%	DB	-13.3	-8.5	2.9	13.6	22.7	28.7	29.2	27.4	21.7	10.6	-2.6	-10.4
		MCWB	-14.3	-10.0	-1.0	5.8	12.1	19.0	20.2	19.0	14.1	5.2	-4.0	-11.7
	5%	DB	-15.3	-11.4	-0.2	10.7	19.9	26.7	27.3	25.3	19.0	7.7	-4.4	-13.7
		MCWB	-16.2	-12.9	-3.2	4.0	10.9	17.7	19.3	18.3	12.8	3.2	-5.7	-14.7
	10%	DB	-17.5	-13.7	-2.4	8.2	17.3	24.8	25.5	23.0	16.4	5.5	-6.5	-17.1
		MCWB	-18.2	-14.8	-4.9	2.6	9.5	16.8	19.0	17.4	11.1	1.8	-7.7	-17.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.1	-7.6	1.4	8.8	15.4	22.5	22.8	22.0	17.4	8.3	-2.3	-9.3
		MCDB	-10.8	-5.5	5.8	16.6	23.2	27.8	27.3	26.6	23.5	13.3	-1.4	-7.7
	2%	WB	-14.3	-10.0	-1.0	6.3	13.4	20.5	21.7	20.5	15.3	5.8	-3.9	-11.6
		MCDB	-13.3	-8.7	2.6	12.8	19.9	26.1	26.3	24.8	19.6	9.7	-2.6	-10.4
	5%	WB	-16.2	-12.7	-3.0	4.3	11.9	19.1	20.8	19.3	13.6	3.6	-5.6	-14.8
		MCDB	-15.4	-11.5	-0.5	9.9	17.8	24.9	25.3	23.3	17.3	7.3	-4.4	-13.8
	10%	WB	-18.1	-14.8	-4.8	2.9	10.4	17.7	19.9	18.1	12.1	1.8	-7.7	-17.8
		MCDB	-17.5	-13.7	-2.5	7.5	16.2	23.1	23.8	22.0	16.0	5.2	-6.6	-17.2
Mean Daily Temperature Range	5% DB	MDBR	13.1	16.3	16.7	12.7	13.6	14.4	12.2	13.0	13.9	12.2	12.7	11.5
		MCDBR	13.0	18.9	18.3	16.7	19.2	18.9	16.6	17.6	18.2	15.2	12.6	11.1
		MCWBR	12.1	17.3	14.6	9.5	9.9	9.7	8.5	9.7	10.8	9.8	10.8	10.1
	5% WB	MCDBR	12.9	18.8	17.4	15.2	16.0	16.3	12.7	14.1	15.1	14.2	12.3	11.3
		MCWBR	12.1	17.3	14.0	8.9	9.1	9.2	7.1	8.3	9.7	9.4	10.5	10.3
Clear-Sky Solar Irradiance	taub		0.247	0.277	0.326	0.455	0.525	0.482	0.459	0.404	0.357	0.330	0.277	0.249
	taud		2.142	2.113	2.040	1.862	1.826	2.049	2.171	2.317	2.388	2.286	2.147	2.073
	Ebn at Noon		713	808	837	760	729	773	786	813	811	738	652	608
	Edh at Noon		64	94	128	177	195	157	137	112	92	80	64	56
All-Sky Solar Radiation	RadAvg		1.19	2.34	4.07	5.28	5.18	5.92	5.16	4.49	3.40	2.20	1.31	0.88
	RadStd		0.05	0.07	0.16	0.38	0.36	0.53	0.49	0.45	0.28	0.13	0.07	0.04

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page